

1 Constructing high-precision tanh networks

We establish three guarantees for a prescribed tanh dictionary: explicit approximation, finite-data recovery, and numerical certification. Error is measured only on $\Omega = [-1, 1]$; halo neurons correct boundary effects without a halo prediction objective.

Targets and geometry. Let $I_\Delta = [-1 - \Delta, 1 + \Delta]$, with $\delta, \Delta, B > 0$. The class $\mathcal{A}_{\delta,B}(I_\Delta)$ consists of functions holomorphic on $\{z : \text{dist}(z, I_\Delta) < \delta\}$, bounded there by B , and real valued on I_Δ . No periodicity or global Fourier-decay condition is imposed. Fix $0 < \lambda \leq 1$ and a halo-design constant $a_H > 0$. Define

$$h = 2/N, \quad x_j = -1 + jh, \quad R = \lceil a_H \sqrt{N} \rceil, \quad \gamma = \lambda/h, \quad W = N + 2R + 1, \quad (1)$$

$$q_c(x) = b + \sum_{j=-R}^{N+R} w_j \tanh(\gamma(x - x_j)), \quad \|c\|_1 = |b| + \sum_j |w_j|.$$

Here N counts core cells and W includes the halo. Smaller λ means wider features relative to the spacing. A width is admissible when

$$(R + \tfrac{1}{2})h \leq \Delta, \quad \gamma\delta \geq 4\pi, \quad \lambda R \geq 2 \log 2, \quad N \geq 16 \max\{1, a_H^2\}. \quad (2)$$

These sufficient conditions hold for all $N \geq N_0$, with explicit N_0 in (C.3).

1.1 Explicit representations with predictable width scaling

Theorem 1 (Boundary-corrected tanh approximation). *For every $f \in \mathcal{A}_{\delta,B}(I_\Delta)$ and admissible width, an explicitly specified real readout c_* satisfies*

$$\|f - q_{c_*}\|_{L^\infty(\Omega)} \leq \frac{128\pi B}{\lambda\delta} \left[\underbrace{e^{-c_{\text{app}}W}}_{\text{resolution and boundary}} + \underbrace{\frac{e^{-\pi^2/\lambda}}{W}}_{\text{grid replicas}} \right] =: E_{\text{rep}}(W, \lambda), \quad (3)$$

$$\|c_*\|_1 \leq KB, \quad c_{\text{app}} = \frac{1}{8} \min\{\pi\delta, \lambda a_H^2\}. \quad (4)$$

The constant $K = K(\lambda, \delta, \Delta)$ is explicit in (B.25) and independent of width and the individual target. Only the readout depends on f .

Correcting the boundary functional with existing halo neurons changes the boundary bound from $e^{-c\lambda R}$ to $he^{-c\lambda R^2}$. Thus $R = O(\sqrt{N})$ suffices, $W = N + O(\sqrt{N})$, and $\gamma = \Theta(W)$. Interior readouts retain the explicit behavior $w_j = hf'(x_j)/2 + O_{\lambda,\delta}(Bh^3)$.

The remaining replica term is explicit, not absorbed into a geometric rate. Readout amplification is uniform in width but can grow as λ decreases: $K = O_{\delta,\Delta}(\lambda^{-3/2}e^{5\pi^2/(24\lambda)})$ is a conservative bound. Appendices A–C give all coefficient formulas and sharper constants.

1.2 Recovery from finitely many real observations

Let $s \geq 2$ be an integer and observe $y_i = f(z_i) + e_i$, with $|e_i| \leq \eta$, at

$$z_i = -1 + ih/s, \quad i = 0, \dots, sN, \quad M_X = sN + 1. \quad (5)$$

Here s counts observation subintervals per center-grid cell. Let A include the bias column, so $(Ac)_i = q_c(z_i)$. All labels lie on the core.

Theorem 2 (Finite-sample recovery with bounded readout). *Under Theorem 1, the constrained fit*

$$c_X \in \arg \min_{\|c\|_1 \leq KB} \|Ac - y\|_\infty \quad (6)$$

admits a solution, and every minimizer satisfies

$$\|f - q_{c_X}\|_{L^\infty(\Omega)} \leq \underbrace{2E_{\text{rep}}(W, \lambda)}_{\text{representation}} + \underbrace{4\eta}_{\text{data error}} + \underbrace{12(1+K)B 2^{-\sqrt{s/2}}}_{\text{finite sampling}}. \quad (7)$$

A feasible fit with maximum-residual objective gap at most η_{opt} adds $2\eta_{\text{opt}}$ to this bound.

For $0 < \tau_s < 1$, the sufficient choice

$$s \geq \max \left\{ 2, \left\lceil 2 \left\lceil \log_2 \frac{12(1+K)}{\tau_s} \right\rceil^2 \right\rceil \right\} \quad (8)$$

makes the sampling contribution at most $B\tau_s$: $M_X = O(W \log^2((1+K)/\tau_s))$, and hence $O(W)$ at fixed budget. The sufficient multiplier is not claimed minimal.

Theorem 2 guarantees constrained maximum-residual fitting. An ordinary least-squares output instead receives the following certificate from its actual residual and readout norm; it need not recover the same coefficients as c_* .

1.3 Certification of the computed network

Let $q_{\text{st}}(x) = \bar{b} + \sum_j \bar{w}_j \tanh(\bar{\gamma}_j x + \bar{\beta}_j)$ be the exact stored network. Let $T_{\text{st}} \geq |\bar{b}| + \sum_j |\bar{w}_j|$ and $r_{\text{cert}} \geq \max_i |q_{\text{st}}(z_i) - y_i|$ be validated bounds. The format has unit roundoff $u \leq 1/4$, half-minimum-subnormal ν , gradual underflow, and no overflow. Use fused affine arguments, correctly rounded tanh and products, and exact accumulation followed by one rounding.

Theorem 3 (A posteriori numerical certificate). *For the grid (5), assume $\max_j |\bar{\gamma}_j| h \leq 2\pi/3$. The computed output $\hat{q}(x)$, including rounding of a real input x to the format, satisfies*

$$\|f - \hat{q}\|_{L^\infty(\Omega)} \leq \underbrace{2(r_{\text{cert}} + \eta)}_{\text{fit and data}} + \underbrace{12(B + T_{\text{st}})2^{-\sqrt{s/2}}}_{\text{between-sample error}} + \underbrace{5uT_{\text{st}}}_{\text{evaluation roundoff}} + \underbrace{\mathcal{U}}_{\text{input and underflow}}, \quad (9)$$

where

$$\mathcal{U} = \frac{B}{\delta}(u + \nu) + [2T_{\text{st}} + (1+u)W + 1]\nu. \quad (10)$$

The certificate applies to any coefficient-recovery method, including least squares.

Appendix E supplies the arithmetic proofs and a certified recovery procedure. Other evaluators require their own error bounds; working precision for fitting may exceed output precision. No small raw-matrix condition number or success of every default LS call is assumed.

Geometric convergence above a prescribed floor. For widths $W \geq W_0$, the replica contribution is bounded by $B\tau_a$, where $\tau_a = 128\pi e^{-\pi^2/\lambda}/(\lambda\delta W_0)$. With $C_{\text{rate}} = 256\pi/(\lambda\delta)$, fixed budgets give

$$\|f - \hat{q}_W\|_\infty \leq C_{\text{rate}} B e^{-c_{\text{app}} W} + \tau, \quad W \geq \frac{1}{c_{\text{app}}} \log \frac{C_{\text{rate}} B}{\varepsilon - \tau}, \quad \varepsilon > \tau. \quad (11)$$

The complete remainder τ is given in (E.15). The logarithmic width statement holds above that fixed budget, not as a joint arbitrary-accuracy theorem when the budget itself tends to zero.

A Local analytic representation and meromorphic quadrature

The following proofs keep all target evaluations inside $U_\delta(I_\Delta)$. The local rectangle and boundary construction consolidate the preceding project note ([Continuous MLPs Project, 2026a](#)); the fixed- λ readout bound and the real-sample and arithmetic results in the subsequent appendices extend it. The finite-interval contour method is related to [Javed and Trefethen \(2014\)](#), but the signed meromorphic formula and its tanh specialization are proved here. No estimates from the earlier global Fourier construction are assumed. In particular, the representation, sampled reconstruction, and computed output are different objects.

A.1 Notation and local coefficient density

For the geometry (1), put

$$d = \frac{\pi}{2\gamma}, \quad r = (R + \frac{1}{2})h, \quad J = [a, b] = [-1 - r, 1 + r], \quad x_0 = -1. \quad (\text{A.1})$$

The W centers are exactly the midpoints of the cells of width h partitioning J ; in particular $Wh = 2 + 2r$. Initially we assume only $r \leq \Delta$ and $d \leq \delta/8$; the stronger conditions in the main text are used when simplifying the rates. The normalized derivative kernel and the coefficient density are

$$K_\gamma(t) = \frac{\gamma}{2} \operatorname{sech}^2(\gamma t), \quad a_\gamma(z) = \frac{f(z + id) - f(z - id)}{2id} = \frac{1}{2d} \int_{-d}^d f'(z + it) dt. \quad (\text{A.2})$$

The second equality follows by integration along the vertical segment.

Lemma A.1 (Local density bound). *If $\sigma + d < \delta$, then a_γ is holomorphic on a neighborhood of $J + i[-\sigma, \sigma]$ and*

$$\sup_{t \in J, |y| \leq \sigma} |a_\gamma(t + iy)| \leq M_\sigma := \frac{B}{\delta - \sigma - d}. \quad (\text{A.3})$$

It is real on J , and $\|a_\gamma\|_{L^\infty(J)} \leq 2B/\delta$.

Proof. Every point $t + iy + iv$, with $t \in J$, $|y| \leq \sigma$, and $|v| \leq d$, has distance at most $\sigma + d$ from I_Δ . Disks about that point of radius less than $\delta - \sigma - d$ lie in the given tube. Cauchy's derivative estimate, followed by a limit in the radius, gives $|f'(t + iy + iv)| \leq M_\sigma$. Average in (A.2). Holomorphy follows from the shifted-value formula. The identity theorem applied to $\overline{f(\bar{z})}$ on the symmetric connected tube gives $f(\bar{z}) = \overline{f(z)}$, so a_γ is real on J . Finally $B/(\delta - d) \leq 2B/\delta$. $\square$

A.2 An exact identity on a finite rectangle

Define

$$D_x(z) = \frac{1}{2} [\tanh(\gamma(x - z)) - \tanh(\gamma(x_0 - z))], \quad F_x(z) = a_\gamma(z) D_x(z), \quad (\text{A.4})$$

$$(C_h f)(x) = f(x_0) + \int_a^b F_x(t) dt. \quad (\text{A.5})$$

Let

$$H_x(z) = \coth(\gamma(x - z)) - \coth(\gamma(x_0 - z)), \quad (\text{A.6})$$

$$(\mathcal{B}_h f)(x) = \frac{1}{4d} \int_{-d}^d [f(a + iy) H_x(a + iy) - f(b + iy) H_x(b + iy)] dy. \quad (\text{A.7})$$

Lemma A.2 (Finite local representation). *For $x \in \Omega$,*

$$f(x) = C_h f(x) + \mathcal{B}_h f(x), \quad (\text{A.8})$$

and

$$\|\mathcal{B}_h f\|_{L^\infty(\Omega)} \leq B [\coth(\gamma r) - \coth(\gamma(r + 2))] \leq \frac{2B}{e^{2\gamma r} - 1}. \quad (\text{A.9})$$

Proof. At $x = x_0$ all differences vanish. For $x > x_0$, $\gamma d = \pi/2$ implies $H_x(t \pm id) = 2D_x(t)$. Therefore

$$\int_a^b F_x(t) dt = \frac{1}{4id} \left[\int_a^b f(t + id) H_x(t + id) dt - \int_a^b f(t - id) H_x(t - id) dt \right]. \quad (\text{A.10})$$

On the counterclockwise boundary of $J + i[-d, d]$, the only poles of fH_x are x and x_0 , with residues $-f(x)/\gamma$ and $f(x_0)/\gamma$. Their integral is $-4id[f(x) - f(x_0)]$. The horizontal part is minus $4id$ times (A.10); the vertical part is $i \int_{-d}^d [f(b + iy) H_x(b + iy) - f(a + iy) H_x(a + iy)] dy$. Division by $-4id$ proves (A.8).

For the bound, set $\ell = x - x_0 \in [0, 2]$. The absolutely convergent expansion $\coth w = 1 + 2 \sum_{k \geq 1} e^{-2kw}$ for $\Re w > 0$, and oddness for $\Re w < 0$, give

$$\begin{aligned} |H_x(a + iy)| &\leq \coth(\gamma r) - \coth(\gamma(r + \ell)), \\ |H_x(b + iy)| &\leq \coth(\gamma(r + 2 - \ell)) - \coth(\gamma(r + 2)). \end{aligned}$$

Each right side is at most $D_r = \coth(\gamma r) - \coth(\gamma(r + 2))$. Thus (A.7) is bounded by $(4d)^{-1}(2d)(2BD_r) = BD_r$. Finally $D_r \leq \coth(\gamma r) - 1 = 2/(e^{2\gamma r} - 1)$. $\square$

The boundary functional is not an unavailable infinite tail of f . It depends only on the two local vertical segments. Discarding it produces (A.9); approximating it instead will improve the halo cost.

A.3 A meromorphic midpoint formula with its endpoint terms

For this subsection only, let $b = a + Mh$ and $t_j = a + (j + 1/2)h$, $0 \leq j < M$. Write $Q_h F = h \sum_j F(t_j)$ and $IF = \int_a^b F(t) dt$. Suppose F is meromorphic on a neighborhood of $[a, b] + i[-\sigma, \sigma]$, with only simple poles in its interior and no poles on the boundary or on $[a, b]$. Denote upper and lower pole sets by $\mathcal{P}_+$ and $\mathcal{P}_-$. Removable poles may be listed with zero residue. Set

$$\zeta_0 = a + h/2, \quad \mathcal{K}_h(z) = \frac{1}{2i} \cot\left(\frac{\pi(z - \zeta_0)}{h}\right), \quad R_h^\pm(z) = \mathcal{K}_h(z) \pm \frac{1}{2}. \quad (\text{A.11})$$

Define

$$\mathcal{H}_h[F] = \int_a^b [R_h^-(t - i\sigma)F(t - i\sigma) - R_h^+(t + i\sigma)F(t + i\sigma)] dt, \quad (\text{A.12})$$

$$\mathcal{V}_h[F] = i \int_0^\sigma \frac{F(b + iy) - F(a + iy) - F(b - iy) + F(a - iy)}{1 + e^{2\pi y/h}} dy, \quad (\text{A.13})$$

$$\mathcal{P}_h[F] = 2\pi i \left[\sum_{p \in \mathcal{P}_+} R_h^+(p) \text{Res}(F; p) + \sum_{p \in \mathcal{P}_-} R_h^-(p) \text{Res}(F; p) \right]. \quad (\text{A.14})$$

Lemma A.3 (Signed meromorphic midpoint identity). *With the stated orientations,*

$$Q_h F - IF = \mathcal{H}_h[F] + \mathcal{V}_h[F] - \mathcal{P}_h[F]. \quad (\text{A.15})$$

Moreover, for $\Im z > 0$ and $\Im w < 0$ respectively,

$$|R_h^+(z)| \leq \frac{1}{e^{2\pi \Im z/h} - 1}, \quad |R_h^-(w)| \leq \frac{1}{e^{2\pi |\Im w|/h} - 1}. \quad (\text{A.16})$$

Proof. Let Γ_+ be $b \rightarrow b + i\sigma \rightarrow a + i\sigma \rightarrow a$ and Γ_- be $a \rightarrow a - i\sigma \rightarrow b - i\sigma \rightarrow b$. Together they traverse the full rectangle counterclockwise. Put $S_\pm = \sum_{p \in \mathcal{P}_\pm} \text{Res}(F; p)$. The residue theorem on the half-rectangles gives

$$\int_{\Gamma_+} F = 2\pi i S_+ - IF, \quad \int_{\Gamma_-} F = 2\pi i S_- + IF. \quad (\text{A.17})$$

Each grid pole of $\mathcal{K}_h$ has residue $h/(2\pi i)$, so on the full rectangle

$$\int_{\Gamma_+ \cup \Gamma_-} \mathcal{K}_h F = Q_h F + 2\pi i \sum_{p \in \mathcal{P}_+ \cup \mathcal{P}_-} \mathcal{K}_h(p) \text{Res}(F; p). \quad (\text{A.18})$$

Substitute $\mathcal{K}_h = R_h^+ - 1/2$ on Γ_+ and $\mathcal{K}_h = R_h^- + 1/2$ on Γ_- . Equations (A.17)–(A.18) give

$$Q_h F - IF = \int_{\Gamma_+} R_h^+ F + \int_{\Gamma_-} R_h^- F - \mathcal{P}_h[F].$$

The horizontal integrals are (A.12). Half-cell endpoint placement yields, for $t = a, b$ and $y \geq 0$,

$$R_h^+(t + iy) = \frac{1}{1 + e^{2\pi y/h}}, \quad R_h^-(t - iy) = -\frac{1}{1 + e^{2\pi y/h}}. \quad (\text{A.19})$$

The four oriented vertical integrals are therefore (A.13). No principal value is required: only F , which has no real poles, was integrated on the real segment in (A.17).

Finally let $w = e^{2\pi i(z - \zeta_0)/h}$. Algebra gives $R_h^+(z) = w/(w - 1)$ and $R_h^-(z) = 1/(w - 1)$. For positive imaginary part, $|w| < 1$ and $|1 - w| \geq 1 - |w|$; for negative imaginary part use $|w - 1| \geq |w| - 1$. This proves (A.16). $\square$

Both endpoint and pole terms are necessary. For example, the entire integrand $F(t) = t^2$ has midpoint error $-(b - a)h^2/12$. Local analyticity alone does not supply exponential finite-interval quadrature; the anchored kernel will make the endpoint integrands small. Likewise, moving a contour past the tanh poles without including $\mathcal{P}_h$ would be invalid.

A.4 Explicit quadrature bounds for the tanh construction

Return to (A.1) and choose

$$k = \left\lceil \frac{\delta}{8d} \right\rceil, \quad \sigma = 2kd. \quad \delta/4 \leq \sigma \leq \delta/2, \quad \sigma + d \leq 5\delta/8, \quad M_\sigma \leq 4B/\delta. \quad (\text{A.20})$$

Since $\gamma\sigma = k\pi$, imaginary periodicity gives $D_x(t \pm i\sigma) = D_x(t)$, while the contour avoids the odd pole layers.

Lemma A.4 (Horizontal, pole, and endpoint contributions). *Uniformly on the core,*

$$|\mathcal{H}_h[F_x]| \leq \frac{4M_\sigma}{e^{2\pi\sigma/h} - 1}, \quad (\text{A.21})$$

$$|\mathcal{P}_h[F_x]| \leq \frac{4\pi M_\sigma}{\gamma} \sum_{\ell=0}^{k-1} \frac{1}{e^{(2\ell+1)\pi^2/\lambda} - 1}, \quad (\text{A.22})$$

$$|\mathcal{V}_h[F_x]| \leq \frac{4M_\sigma h \log 2}{\pi(e^{2\gamma r} - 1)}. \quad (\text{A.23})$$

Proof. For real t , $D_x(t) = \int_{x_0}^x K_\gamma(v - t)dv \geq 0$ and $\int_{\mathbb{R}} D_x(t)dt = x - x_0 \leq 2$, since K_γ has unit mass. Thus $\int_j |F_x(t \pm i\sigma)|dt \leq 2M_\sigma$. Apply (A.16) to obtain (A.21).

If $x > x_0$, the possible poles are $x \pm i(2\ell + 1)d$ and $x_0 \pm i(2\ell + 1)d$, $0 \leq \ell < k$. Their residues in D_x are respectively $-1/(2\gamma)$ and $1/(2\gamma)$. Each residue of F_x has modulus at most $M_\sigma/(2\gamma)$, and there are at most four at each absolute height. Because $2\pi(2\ell + 1)d/h = (2\ell + 1)\pi^2/\lambda$, (A.14) implies (A.22). For $x = x_0$ the integrand is identically zero.

On either vertical endpoint, subtract the common saturated value of the two tanhs. The identity $\tanh w - 1 = -2/(e^{2w} + 1)$ for $\Re w > 0$, and oddness for $\Re w < 0$, give $|D_x(a \pm iy)|, |D_x(b \pm iy)| \leq 2/(e^{2\gamma r} - 1)$. Use the four endpoint terms in (A.13) and $\int_0^\infty (1 + e^{2\pi y/h})^{-1} dy = h \log 2/(2\pi)$. $\square$

To keep the residue series distinct from its elementary envelope, define

$$\mathcal{S}(A) = \sum_{\ell \geq 0} \frac{1}{e^{(2\ell+1)A} - 1}, \quad \mathcal{U}(A) = \frac{e^{-A}}{(1 - e^{-A})(1 - e^{-2A})}. \quad (\text{A.24})$$

For $A > 0$,

$$\mathcal{S}(A) \leq \frac{1}{1 - e^{-A}} \sum_{\ell \geq 0} e^{-(2\ell+1)A} = \mathcal{U}(A). \quad (\text{A.25})$$

Only this scalar bound is extended to infinitely many layers; no target value is used outside the finite contour. For $A \geq \pi^2$, $\mathcal{U}(A) \leq 2e^{-A}$.

Proposition A.5 (Baseline network and signed error). *The explicit network*

$$q_h(x) = f(x_0) + h \sum_{j=-R}^{N+R} a_\gamma(x_j) D_x(x_j) \quad (\text{A.26})$$

satisfies

$$f - q_h = \mathcal{B}_h f - \mathcal{V}_h[F_x] + \mathcal{P}_h[F_x] - \mathcal{H}_h[F_x]. \quad (\text{A.27})$$

Its baseline readout obeys

$$|w_j^{\text{base}}| \leq Bh/\delta, \quad \sum_j |w_j^{\text{base}}| \leq 2B(1 + \Delta)/\delta, \quad |b^{\text{base}}| \leq B + 2B(1 + \Delta)/\delta. \quad (\text{A.28})$$

The uncorrected error is at most

$$\frac{16B}{\delta(e^{\pi\delta/(2h)} - 1)} + \frac{16\pi Bh}{\lambda\delta} \mathcal{U}(\pi^2/\lambda) + \frac{B(2 + 16h \log 2/(\pi\delta))}{e^{2\lambda(R+1/2)} - 1}. \quad (\text{A.29})$$

Proof. Subtract the quadrature identity from (A.8) to get (A.27); combine the preceding bounds for (A.29). The weights are $ha_\gamma(x_j)/2$. Lemma A.1, $Wh = 2 + 2r \leq 2 + 2\Delta$, and the anchored bias give (A.28). $\square$

B Boundary correction and width-independent readout control

The correction must remove the combined functional

$$\mathcal{B}_h^{\text{eff}} f(x) = \mathcal{B}_h f(x) - \mathcal{V}_h[F_x], \quad f - q_h = \mathcal{B}_h^{\text{eff}} f + \mathcal{P}_h[F_x] - \mathcal{H}_h[F_x]. \quad (\text{B.1})$$

The first term is the finite-representation boundary contribution; the second is the midpoint endpoint contribution. Correcting only the first would not give the final estimate.

B.1 Local boundary series and the improved moment estimate

Set

$$z_-(x) = e^{-2\gamma(x-a)}, \quad z_+(x) = e^{-2\gamma(b-x)}, \quad \theta = e^{-\gamma r}, \quad z_{\pm,0} = z_{\pm}(x_0). \quad (\text{B.2})$$

On the core all these arguments lie in $(0, \theta^2]$. For integers $\ell \geq 1$, define the finite local moments

$$\mu_\ell^- = \frac{1}{2d} \int_{-d}^d f(a + iy) e^{2i\ell\gamma y} dy, \quad \mu_\ell^+ = \frac{1}{2d} \int_{-d}^d f(b + iy) e^{-2i\ell\gamma y} dy, \quad (\text{B.3})$$

$$v_\ell^- = i(-1)^\ell \int_0^\sigma \frac{a_\gamma(a - iy) e^{-2i\ell\gamma y} - a_\gamma(a + iy) e^{2i\ell\gamma y}}{1 + e^{2\pi y/h}} dy, \quad (\text{B.4})$$

$$v_\ell^+ = i(-1)^\ell \int_0^\sigma \frac{a_\gamma(b - iy) e^{2i\ell\gamma y} - a_\gamma(b + iy) e^{-2i\ell\gamma y}}{1 + e^{2\pi y/h}} dy. \quad (\text{B.5})$$

Put

$$g_\ell^\pm = \mu_\ell^\pm - \nu_\ell^\pm, \quad G_\pm(z) = \sum_{\ell \geq 1} g_\ell^\pm z^\ell, \quad D_\lambda = \frac{\pi}{2\lambda} + \frac{4 \log 2}{\pi}. \quad (\text{B.6})$$

Lemma B.1 (Boundary series with $O(h)$ coefficients). *The coefficients $g_\ell^\pm$ are real and obey*

$$|g_\ell^\pm| \leq M_h := \frac{Bh}{\delta} D_\lambda. \quad (\text{B.7})$$

The functions $G_\pm$ are holomorphic on $|z| < 1$, and

$$\mathcal{B}_h^{\text{eff}} f(x) = G_-(z_-(x)) - G_-(z_{-,0}) + G_+(z_+(x)) - G_+(z_{+,0}), \quad x \in \Omega. \quad (\text{B.8})$$

Proof. For $\Re w > 0$, $\coth w = 1 + 2 \sum_{\ell \geq 1} e^{-2\ell w}$ and $\tanh w = 1 + 2 \sum_{\ell \geq 1} (-1)^\ell e^{-2\ell w}$. Use oddness at the right boundary. With $\Delta z_\pm^\ell = z_\pm(x)^\ell - z_{\pm,0}^\ell$, this gives

$$\begin{aligned} H_x(a + iy) &= 2 \sum_{\ell \geq 1} \Delta z_-^\ell e^{2i\ell\gamma y}, & H_x(b + iy) &= -2 \sum_{\ell \geq 1} \Delta z_+^\ell e^{-2i\ell\gamma y}, \\ D_x(a + iy) &= \sum_{\ell \geq 1} (-1)^\ell \Delta z_-^\ell e^{2i\ell\gamma y}, & D_x(b + iy) &= - \sum_{\ell \geq 1} (-1)^\ell \Delta z_+^\ell e^{-2i\ell\gamma y}. \end{aligned}$$

The series converge uniformly on the finite integration segments because all real boundary coordinates are at most $\theta^2 < 1$. Termwise integration in (A.7) and (A.13) yields (B.3)–(B.8) with the displayed signs.

For the sharper bound, $\gamma d = \pi/2$ gives $\int_{-d}^d e^{2i\ell\gamma y} dy = 0$ for every $\ell \geq 1$. Subtract $f(a)$ in μ_ℓ^- and $f(b)$ in μ_ℓ^+ . Cauchy's estimate implies $|f'(a + it)|, |f'(b + it)| \leq 2B/\delta$ for $|t| \leq d$. Hence

$$|\mu_\ell^\pm| \leq \frac{1}{2d} \frac{2B}{\delta} \int_{-d}^d |y| dy = \frac{Bd}{\delta} = \frac{\pi Bh}{2\lambda\delta}.$$

Also $|\nu_\ell^\pm| \leq 2M_\sigma \int_0^\sigma (1 + e^{2\pi y/h})^{-1} dy \leq 4Bh \log 2/(\pi\delta)$. This proves (B.7). The positive and negative y integrands in μ are conjugates, and the bracket in each ν is a conjugate minus the original number. Multiplication by i makes it real. Finally bounded coefficients imply holomorphy of $G_\pm$ in the unit disk, with $|G_\pm(z)| \leq M_h |z|/(1 - |z|)$. $\square$

The cancellation in $\mu_\ell^\pm$ is the source of the extra factor h . No target value outside the local tube is used, and only $g_1^\pm, \dots, g_m^\pm$ will enter the finite correction.

B.2 Rational correction using the existing outer-halo neurons

Lemma B.2 (Approximation in a prescribed rational denominator). *Let $0 < \theta < 1$, and let $G(z) = \sum_{\ell \geq 1} g_\ell z^\ell$ with $|g_\ell| \leq M$. Choose distinct $0 < t_i \leq \theta^{-1}$, $1 \leq i \leq m$. Define*

$$D_m(z) = \prod_{i=1}^m (1 + t_i z), \quad P_m = [D_m G]_{\leq m}, \quad S_m = P_m / D_m, \quad (\text{B.9})$$

where $[\cdot]_{\leq m}$ means Taylor truncation through degree m . Then

$$\sup_{0 \leq z \leq \theta^2} |G(z) - S_m(z)| \leq \frac{M\theta}{(1 - \theta)^2} (2\theta)^m. \quad (\text{B.10})$$

Moreover $S_m = c_0 + \sum_{i=1}^m c_i/(1 + t_i z)$, with real coefficients if g_ℓ, t_i are real.

Proof. On $|z| = \theta$, $|G(z)| \leq M/(1 - \theta)$ and $|D_m(z)| \leq \prod_{i=1}^m (1 + t_i \theta) \leq 2^m$. Cauchy's coefficient estimate gives $|[z^k] D_m G| \leq M 2^m \theta^{-k}/(1 - \theta)$. For $0 \leq z \leq \theta^2$, the tail beyond degree m is at most $M 2^m \theta^{m+1}/(1 - \theta)^2$. Since $D_m(z) \geq 1$ on the positive real axis, division proves (B.10). Polynomial division and partial fractions apply because $\deg P_m \leq \deg D_m = m$ and its denominator roots are distinct. $\square$

Choose any $1 \leq m \leq \lfloor (R+1)/2 \rfloor$ and the m outermost centers on each side:

$$\xi_i^- = a + (i - \frac{1}{2})h = x_{-R+i-1}, \quad \xi_i^+ = b - (i - \frac{1}{2})h = x_{N+R-i+1}, \quad t_i = e^{2\lambda(i-1/2)}. \quad (\text{B.11})$$

These centers lie in the outer halves of the halos, so $(i - 1/2)h \leq r/2$ and $t_i \leq e^{\gamma r} = \theta^{-1}$. The feature identities are

$$\frac{1}{1 + t_i z_-(x)} = \frac{1 + \tanh(\gamma(x - \xi_i^-))}{2}, \quad \frac{1}{1 + t_i z_+(x)} = \frac{1 - \tanh(\gamma(x - \xi_i^+))}{2}. \quad (\text{B.12})$$

Apply Lemma B.2 separately to G_- and G_+ and set

$$q_{\text{bd}}(x) = S_m^-(z_-(x)) - S_m^-(z_{-,0}) + S_m^+(z_+(x)) - S_m^+(z_{+,0}). \quad (\text{B.13})$$

Proposition B.3 (Correction in the same hidden dictionary). *The network $q_* = q_h + q_{\text{bd}}$ has the same W hidden-unit slots as q_h , is real valued, and satisfies*

$$\|\mathcal{B}_h^{\text{eff}} f - q_{\text{bd}}\|_{L^\infty(\Omega)} \leq \frac{4M_h\theta}{(1-\theta)^2} (2\theta)^m. \quad (\text{B.14})$$

If $m = \lfloor (R+1)/2 \rfloor$ and $\lambda R \geq 2 \log 2$, then

$$\|\mathcal{B}_h^{\text{eff}} f - q_{\text{bd}}\|_\infty \leq 16M_h e^{-\lambda R^2/4}. \quad (\text{B.15})$$

Proof. Apply (B.10) at the four arguments in the two anchored differences. Identity (B.12) realizes the correction in the chosen tanh features; constants in each anchored difference cancel. For (B.15), $m \geq R/2$, $\theta \leq e^{-\lambda R} \leq 1/4$, and

$$(2\theta)^m = e^{-m(\lambda(R+1/2) - \log 2)} \leq e^{-(R/2)(\lambda R/2)} = e^{-\lambda R^2/4}.$$

Also $4\theta/(1-\theta)^2 \leq 16$. This proves the stated conservative constant. $\square$

For direct export, write $D_m(z) = \sum_{j=0}^m d_j z^j$, set $g_0^\pm = 0$, and compute

$$P_m^\pm(z) = \sum_{k=0}^m \left(\sum_{j=0}^k d_j g_{k-j}^\pm \right) z^k, \quad c_i^\pm = \frac{P_m^\pm(-1/t_i)}{\prod_{j \neq i} (1 - t_j/t_i)}. \quad (\text{B.16})$$

The actual correction is

$$\begin{aligned} q_{\text{bd}}(x) &= \frac{1}{2} \sum_{i=1}^m c_i^- [\tanh(\gamma(x - \xi_i^-)) - \tanh(\gamma(x_0 - \xi_i^-))] \\ &\quad - \frac{1}{2} \sum_{i=1}^m c_i^+ [\tanh(\gamma(x - \xi_i^+)) - \tanh(\gamma(x_0 - \xi_i^+))]. \end{aligned} \quad (\text{B.17})$$

Thus increase the left selected weights by $c_i^-/2$, decrease the right weights by $c_i^+/2$, and set the final bias to $f(x_0) - \sum_j w_j \tanh(\gamma(x_0 - x_j))$. No exponential or rational activation is evaluated by the resulting MLP.

B.3 A scaled coefficient formula at fixed relative scale

Direct use of (B.16) obscures cancellation between its numerator and denominator. The following identity estimates both together. Let

$$\zeta = e^{-2\lambda}, \quad r_i = t_i^{-1} = \zeta^{i-1/2}, \quad p_k = \prod_{j=1}^k (1 - \zeta^j), \quad p_0 = 1. \quad (\text{B.18})$$

Here r_i are reciprocal rational nodes, not the physical halo margin r . For a finite list, e_k denotes its elementary symmetric polynomial of degree k , with $e_0 = 1$ and $e_k = 0$ for degrees above the list length.

Lemma B.4 (Exact scaled partial-fraction coefficients). *For either boundary function G , the nonconstant coefficients in (B.16) equal*

$$c_i = (-1)^i \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1}p_{m-i}} \sum_{\ell=1}^m e_{\ell-1}(r_1, \dots, \widehat{r_i}, \dots, r_m) g_\ell. \quad (\text{B.19})$$

The hat omits r_i .

Proof. Factor $D_m = (1 + t_i z) D_{m,i}$ and write $D_{m,i} = \sum_{k=0}^{m-1} d_k^{(i)} z^k$. For $0 \leq L \leq m-1$, evaluation at $z = -r_i = -1/t_i$ gives

$$\sum_{k=0}^L [z^k] D_m (-r_i)^k = \sum_{k=0}^L (d_k^{(i)} + t_i d_{k-1}^{(i)}) (-r_i)^k = d_L^{(i)} (-r_i)^L.$$

Consequently $P_m(-r_i) = (-r_i)^m \sum_{\ell=1}^m g_\ell d_{m-\ell}^{(i)}$. Reversing the elementary symmetric coefficients yields $d_{m-\ell}^{(i)} = (\prod_{j \neq i} t_j) e_{\ell-1}(r_{\neq i})$. In the denominator,

$$\prod_{j \neq i} (1 - t_j/t_i) = (-1)^{m-i} \zeta^{-(m-i)(m-i+1)/2} p_{i-1} p_{m-i}.$$

The numerator's sign is $(-1)^m$ and its power of ζ is $m(i-1/2) - m^2/2 + i - 1/2$. Adding the denominator's compensating power reduces this to $[i(i+1)-1]/2$. Division proves (B.19). $\square$

Define the convergent products and the explicit amplification constant

$$p_\infty = \prod_{j \geq 1} (1 - \zeta^j), \quad L_+ = \prod_{j \geq 1} (1 + \zeta^{j-1/2}), \quad (\text{B.20})$$

$$C_{\text{bd}}(\lambda) = \sum_{i \geq 1} \frac{\zeta^{[i(i+1)-1]/2}}{p_{i-1} p_\infty} \prod_{\substack{j \geq 1 \\ j \neq i}} (1 + \zeta^{j-1/2}). \quad (\text{B.21})$$

Proposition B.5 (Width-independent boundary amplification). *For every fixed $\lambda > 0$, $0 < C_{\text{bd}}(\lambda) < \infty$, and*

$$\sum_{i=1}^m |c_i| \leq C_{\text{bd}}(\lambda) \|g\|_\infty, \quad \|c(g) - c(\widetilde{g})\|_1 \leq C_{\text{bd}}(\lambda) \|g - \widetilde{g}\|_\infty. \quad (\text{B.22})$$

One fully elementary upper bound is

$$C_{\text{bd}}(\lambda) \leq \frac{\sqrt{\pi} e^\lambda}{2\sqrt{\lambda}} \exp\left(\frac{5\pi^2}{24\lambda}\right). \quad (\text{B.23})$$

Proof. All symmetric coefficients in (B.19) are nonnegative, and $\sum_{\ell=1}^m e_{\ell-1}(r_{\neq i}) = \prod_{j \neq i} (1 + r_j)$. Also $p_{m-i} \geq p_\infty > 0$. The absolute row sum is therefore bounded by the i th summand in (B.21). Geometric convergence of $\sum \zeta^j$ implies positivity of p_∞ and finiteness of L_+ , by taking logarithms. The remaining factor decays quadratically in i in its exponent, so the sum is finite. Linearity proves the perturbation bound.

For (B.23), first

$$\log p_\infty^{-1} = \sum_{k \geq 1} \frac{1}{k(e^{2\lambda k} - 1)} \leq \frac{1}{2\lambda} \sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{12\lambda}.$$

The function $F(t) = \log(1 + e^{-t})$ is convex, so its midpoint value on each interval of length 2λ is at most the interval average. Therefore

$$\log L_+ \leq \frac{1}{2\lambda} \int_0^\infty \log(1 + e^{-t}) dt = \frac{\pi^2}{24\lambda}.$$

The integral follows by the absolutely integrable alternating exponential series. Finally

$$\sum_{i \geq 1} \zeta^{[i(i+1)-1]/2} = e^\lambda \sum_{i \geq 1} e^{-\lambda i(i+1)} \leq e^\lambda \int_0^\infty e^{-\lambda t^2} dt = \frac{e^\lambda \sqrt{\pi}}{2\sqrt{\lambda}}.$$

Use $p_{i-1} \geq p_\infty$ and bound each omitted-factor product by L_+ . $\square$

The infinite products define constants, not target information. They can be bounded with finite products and geometric tail bounds; alternatively (B.23) is a closed conservative bound. Formula (B.19) contains no powers $t_i \gg 1$, but its exact sensitivity bound alone is not a floating-point implementation theorem for computing the moments. The real-data recovery below avoids computing those moments numerically.

Proposition B.6 (Corrected readout bound). *For the corrected explicit network,*

$$\sum_j |w_j^*| \leq \frac{2B(1+\Delta)}{\delta} + C_{\text{bd}}(\lambda) \frac{Bh}{\delta} D_\lambda, \quad \|c_*\|_1 \leq B \left[1 + \frac{4(1+\Delta)}{\delta} + \frac{2C_{\text{bd}}(\lambda)D_\lambda h}{\delta} \right]. \quad (\text{B.24})$$

For all main-text admissible widths, a common cap is

$$K(\lambda, \delta, \Delta) = 1 + \frac{4(1+\Delta)}{\delta} + \frac{C_{\text{bd}}(\lambda)D_\lambda}{4\delta}. \quad (\text{B.25})$$

In particular $K = O_{\delta, \Delta}(\lambda^{-3/2} e^{5\pi^2/(24\lambda)})$ as $\lambda \downarrow 0$.

Proof. Each boundary contributes half its partial-fraction absolute sum to the tanh readout. Combine Lemma B.1, Proposition B.5, and (A.28). The anchored bias is at most $B + \sum_j |w_j^*|$. Since admissibility gives $N \geq 16$, $h \leq 1/8$, proving (B.25). The stated small- λ bound follows from (B.23) and $D_\lambda = O(\lambda^{-1})$ on $(0, 1]$. $\square$

Remark B.7 (Interior weights and explicitness). At centers not selected for correction the weights remain $ha_\gamma(x_j)/2$. Taylor's theorem in a disk of radius below δ gives

$$w_j^* = \frac{h}{2} f'(x_j) - \frac{\pi^2 h^3}{48\lambda^2} f'''(x_j) + O\left(\frac{Bh^5}{\lambda^4 \delta^5}\right), \quad (\text{B.26})$$

under $d/\delta \leq 1/8$. For example, the discarded odd Taylor terms in the density are bounded by $(B/\delta)(d/\delta)^4/(1 - (d/\delta)^2)$. All coefficients in (A.2) and (B.3)–(B.17) are prescribed linear functionals of f . Their existence and magnitude do not require solving an unspecified approximation problem. The later sampled algorithm is a different way to recover an accurate readout in the same space.

C Proof of the representation theorem and explicit constants

C.1 The uncompressed approximation bound

Combining (B.1) with the constructed correction yields

$$f - q_* = (\mathcal{B}_h^{\text{eff}} f - q_{\text{bd}}) + \mathcal{P}_h[F_x] - \mathcal{H}_h[F_x]. \quad (\text{C.1})$$

Thus Lemma A.4, (A.25), and Proposition B.3 give the explicit bound

$$\begin{aligned} \|f - q_*\|_\infty &\leq \frac{16B}{\delta(e^{\pi\delta/(2h)} - 1)} + \frac{16\pi Bh}{\lambda\delta} \mathcal{U}(\pi^2/\lambda) \\ &\quad + \frac{4BhD_\lambda}{\delta} \frac{\theta}{(1-\theta)^2} (2\theta)^m, \quad \theta = e^{-\lambda(R+1/2)}. \end{aligned} \quad (\text{C.2})$$

This version is valid for every $\lambda > 0$, $r \leq \Delta$, $d \leq \delta/8$, and $1 \leq m \leq \lfloor (R+1)/2 \rfloor$. A decreasing correction bound further requires $2\theta < 1$. It displays all nonasymptotic denominators and may be used instead of the simpler main-text envelope.

C.2 Admissible widths and the conversion to total hidden width

For fixed $0 < \lambda \leq 1$, $a_H > 0$, and $\delta, \Delta > 0$, one sufficient integer threshold for (2) is

$$N_0 = \left\lceil \max \left\{ 16, 16a_H^2, \frac{16a_H^2}{\Delta^2}, \frac{6}{\Delta}, \frac{8\pi}{\lambda\delta}, \left(\frac{2 \log 2}{\lambda a_H} \right)^2 \right\} \right\rceil. \quad (\text{C.3})$$

Indeed, with $R = \lceil a_H \sqrt{N} \rceil$,

$$r \leq \frac{2a_H}{\sqrt{N}} + \frac{3}{N} \leq \Delta, \quad \gamma\delta = \frac{\lambda\delta N}{2} \geq 4\pi, \quad \lambda R \geq \lambda a_H \sqrt{N} \geq 2 \log 2. \quad (\text{C.4})$$

Also, for the main-text admissible widths,

$$N \leq W = N + 2R + 1 \leq N + 2a_H \sqrt{N} + 3 \leq 2N. \quad (\text{C.5})$$

The last inequality follows from $N \geq 16a_H^2$ and $N \geq 16$. No halo neuron is omitted from the width count.

Proof of Theorem 1. Use the explicit q_* from (A.26) and (B.13), with $m = \lfloor (R+1)/2 \rfloor$. The first term in (C.2) is at most $(32B/\delta)e^{-\pi\delta N/4}$: the exponent is at least $\log 2$ under $\gamma\delta \geq 4\pi$ and $\lambda \leq 1$. By (B.15), the boundary term is at most

$$\frac{32BD\lambda}{\delta N} e^{-\lambda a_H^2 N/4}. \quad (\text{C.6})$$

Set $\kappa = \min\{\pi\delta, \lambda a_H^2\}/4$. The sum of these two terms is bounded by

$$\frac{32B(1+D\lambda)}{\delta} e^{-\kappa N} \leq \frac{128\pi B}{\lambda\delta} e^{-\kappa W/2}. \quad (\text{C.7})$$

Here we used (C.5) and $\lambda(1+D\lambda) \leq 1 + \pi/2 + 4 \log 2/\pi < 4\pi$. For the pole term, $\mathcal{U}(A) \leq 2e^{-A}$ when $A \geq \pi^2$ gives

$$\frac{32\pi B}{\lambda\delta N} \mathcal{U}(\pi^2/\lambda) \leq \frac{64\pi B}{\lambda\delta N} e^{-\pi^2/\lambda} \leq \frac{128\pi B}{\lambda\delta W} e^{-\pi^2/\lambda}. \quad (\text{C.8})$$

Equations (C.7)–(C.8) give the stated $c_{\text{app}} = \kappa/2$ and (3). Proposition B.6 proves the coefficient cap. The coefficient maps are real and linear in f by their construction in Appendix B. $\square$

C.3 Parameter consequences and scope of the rate

The bounds give $R = O(\sqrt{N})$, $W = N + O(\sqrt{N})$, and $\gamma = \lambda N/2 = \Theta(W)$ when the design is fixed. Hidden biases satisfy $|\gamma x_j| \leq \gamma(1+\Delta) = O(W)$, while the target-dependent readout obeys $\|c_*\|_1 \leq KB$. A conventional dense scalar-input/output shallow network has $3W+1$ stored scalar parameters; its hidden geometry is target-independent.

For $W \geq W_0$ the elementary replica term in Theorem 1 is at most $B\tau_a$ from the main text. The sharper expression gives an even smaller certificate: $\gamma\delta \geq 4\pi$ implies

$$E_{\text{pole}} \leq 4B \mathcal{U}(\pi^2/\lambda). \quad (\text{C.9})$$

At $\lambda = 1/4$ its right side is approximately $2.863 \times 10^{-17}B$. This number is an approximation contribution only; B is the complex-neighborhood magnitude bound, not necessarily the real supremum of f . It is not a complete binary64 error guarantee.

Boundary correction is not the sole way to obtain a geometric-to-floor envelope at fixed λ : taking R proportional to N in the uncorrected formula also makes its halo term geometric. The present gain

is the stronger boundary estimate with a sublinear halo count, together with controlled corrected weights. Neither rule removes the nonzero fixed- λ replica coefficient.

The fixed- γ geometric-to-zero construction in [Continuous MLPs Project \(2026a\)](#) is a different parameter schedule, with $\lambda = \gamma h \rightarrow 0$. Its numerical stability is not claimed here. This paper-style statement instead fixes λ and proves a width-refining representation with bounded readout. No assertion of asymptotic geometric convergence to zero, uniformly small raw Gram condition number, or necessity of uniform centers for every accurate MLP is made.

D Sampling and recovery from real core observations

The target’s complex extension is used to establish a bounded approximation witness, not requested as input to the sampled algorithm. In contrast to the buffered least-squares objective considered in the preceding note, the fitting and reporting intervals here are both Ω . The following residual inequality avoids coefficientwise inversion and does not impose a halo prediction objective.

D.1 Two local polynomial facts

Lemma D.1 (Analytic polynomial approximation on one cell). *Let g be holomorphic on a neighborhood of the closed Bernstein ellipse of parameter $\rho > 1$ about $[-1, 1]$, and $|g| \leq M$ there. There is a real degree- p polynomial P_p when g is real on the interval, satisfying*

$$\|g - P_p\|_{L^\infty([-1, 1])} \leq \frac{2M}{\rho - 1} \rho^{-p}. \quad (\text{D.1})$$

The same estimate holds on an affine image of the ellipse and interval.

Proof. Compose g with the Joukowski map $(w + w^{-1})/2$. The resulting Laurent series has equal positive and negative coefficients, by invariance under $w \mapsto 1/w$. Cauchy’s formula on $|w| = \rho$ bounds the nonconstant Chebyshev coefficient by $2M\rho^{-k}$. Truncating through degree p , using $|T_k(x)| \leq 1$ on the interval, and summing $2M \sum_{k>p} \rho^{-k}$ proves the claim. Conjugate symmetry gives real coefficients. An affine change of variable gives the cell version. This is the standard estimate in [Trefethen \(2019, Chapter 8\)](#). $\square$

Lemma D.2 (A sufficient equispaced norming grid). *Let P have degree at most $p \geq 1$ on an interval of length h . If $s + 1$ uniformly spaced points, including endpoints, divide the interval into $s \geq 2p^2$ subintervals, then*

$$\|P\|_\infty \leq 2 \max_{\text{these points}} |P|. \quad (\text{D.2})$$

Proof. The classical Markov inequality is $\|P'\|_{[-1, 1]} \leq p^2 \|P\|_{[-1, 1]}$ (see, for example, the statement recalled by [Aptekarev et al. \(2014\)](#)). Affine scaling gives $\|P'\| \leq (2p^2/h) \|P\|$. A maximizer of $|P|$ lies within $h/(2s)$ of a grid point, hence $\|P\| \leq \max |P| + (p^2/s) \|P\|$. Rearrangement proves the result. $\square$

Only the classical Markov inequality is invoked in Lemma D.2; the residual and tanh-space conclusions below are not statements imported from polynomial interpolation theory.

D.2 A solver-independent sample-to-core bound

For any real network $q(x) = b + \sum_{j=1}^W w_j \tanh(\gamma_j x + \beta_j)$, let $T \geq |b| + \sum_j |w_j|$ and $\Gamma = \max_j |\gamma_j|$. The centers need not be uniform for this subsection; the residual observation cells still have length h .

Lemma D.3 (Local residual certificate). *Assume $\Gamma h \leq 2\pi/3$ and $5h/8 < \delta$. For X as in (5), $s \geq 2$, one has*

$$\|f - q\|_{L^\infty(\Omega)} \leq 2 \max_{z_i \in X} |f(z_i) - q(z_i)| + 12(B + T)2^{-\sqrt{s/2}}. \quad (\text{D.3})$$

More generally, for $\rho > 1$ satisfying

$$\Gamma \frac{h}{4}(\rho - \rho^{-1}) \leq \pi/4, \quad \frac{h}{4}(\rho + \rho^{-1}) < \delta, \quad (\text{D.4})$$

and $s \geq 2\rho^2$, the final term can be replaced by $6(B + T)\rho^{-p}/(\rho - 1)$.

Proof. For real u, v ,

$$|\tanh(u + iv)|^2 = \frac{\sinh^2 u + \sin^2 v}{\sinh^2 u + \cos^2 v} \leq 1 \quad \text{when } |v| \leq \pi/4. \quad (\text{D.5})$$

The network is holomorphic on a neighborhood of that closed strip after scaling, and $|q(z)| \leq T$ when $|\Im z| \leq \pi/(4\Gamma)$. Each core cell has midpoint in I_Δ . Its scaled Bernstein ellipse has semiaxes $h(\rho + \rho^{-1})/4$ and $h(\rho - \rho^{-1})/4$. The first inequality in (D.4) keeps all feature arguments in (D.5); the second keeps the entire ellipse inside the target tube, by distance from its midpoint. Thus $f - q$ is analytic and bounded by $B + T$ on every such ellipse.

On one cell, Lemma D.1 gives a polynomial P of degree p with residual approximation error $e_p = 2(B + T)\rho^{-p}/(\rho - 1)$. Lemma D.2 gives

$$\|f - q\|_{\text{cell}} \leq \|P\| + e_p \leq 2 \max_{X \cap \text{cell}} |f - q| + 3e_p.$$

Maximize over cells for the general claim. For the main statement set $\rho = 2$ and $p = \lfloor \sqrt{s/2} \rfloor \geq 1$. The conditions become exactly those stated in the lemma. Since $2^{-p} \leq 2 \cdot 2^{-\sqrt{s/2}}$, (D.3) follows. $\square$

For the main-text dictionary, $\Gamma h = \lambda \leq 1 < 2\pi/3$. Also $\gamma\delta \geq 4\pi$ implies $\delta N \geq 8\pi/\lambda > 5/4$, so $5h/8 < \delta$. Thus no extra width restriction is required for Theorem 2. The polynomial degree and ellipse are proof devices only. They have been eliminated from the main rate $12(B + T)2^{-\sqrt{s/2}}$ and from the algorithm.

D.3 Proof of the finite-data theorem

Proof of Theorem 2. The coefficient ball $\|c\|_1 \leq KB$ is compact, and the sampled objective is continuous, so a minimizer exists. The readout supplied by Theorem 1 is feasible. Thus, for any feasible η_{opt} -suboptimal candidate,

$$\|Ac_X - y\|_\infty \leq \|Ac_* - y\|_\infty + \eta_{\text{opt}} \leq E_{\text{rep}}(W, \lambda) + \eta + \eta_{\text{opt}}. \quad (\text{D.6})$$

Consequently $\max_i |f(z_i) - q_{c_X}(z_i)| \leq E_{\text{rep}}(W, \lambda) + 2\eta + \eta_{\text{opt}}$. Use Lemma D.3 with $T = KB$. Setting $\eta_{\text{opt}} = 0$ proves the displayed theorem; the general statement adds $2\eta_{\text{opt}}$. Finally (8) is obtained by solving $12(1 + K)2^{-\sqrt{s/2}} \leq \tau_s$ for s . $\square$

There need not be a unique coefficient minimizer. The result is uniform over all minimizers and gives noise-robust recovery on the target class. It is not an assertion that the nonlinear solution map is a unique bounded linear projector. The observation count is exactly $sN + 1 = s(W - 2R - 1) + 1$, not exactly sW . A fixed sampling floor makes s independent of width; varying that floor incurs the explicit quadratic logarithmic factor in (8).

D.4 What ordinary least squares does and does not inherit

With noiseless samples, every exact unconstrained least-squares minimizer satisfies

$$\frac{\|A_{\text{CLS}} - y\|_2}{\sqrt{M_X}} \leq E_{\text{rep}}(W, \lambda), \quad (\text{D.7})$$

because c_* is a comparison vector. This only controls sampled root-mean-square error, not the maximum residual or the readout magnitude. For any LS output, however, Lemma D.3 gives

$$\|f - q_{\text{LS}}\|_\infty \leq 2(r_{\text{LS}} + \eta) + 12(B + T_{\text{LS}})2^{-\sqrt{s/2}}, \quad (\text{D.8})$$

where $r_{\text{LS}} \geq \|A_{\text{CLS}} - y\|_\infty$ and $T_{\text{LS}} \geq \|c_{\text{LS}}\|_1$ are measured or validated for the returned network. No coefficient cap had to be imposed during that solve.

The existence of a bounded witness does not imply $\|c_{\text{LS}}\|_1 \leq KB$: for full column rank, writing $y = Ac_* + e$ gives $c_{\text{LS}} = c_* + A^\dagger e$. Small residual components can be magnified in poorly resolved coefficient directions. The explicit QI coefficient map and a fitted coefficient map are therefore distinct, as already emphasized in the old fixed-dictionary experiment proposal (Continuous MLPs Project, 2026b, Section 8.1). No unconditional least-squares stability theorem is asserted by (D.8).

E Numerical certification and a finite implementation

This appendix distinguishes three quantities: the exact function q_{st} defined by stored parameters, its computed evaluation $\widehat{q}$, and the unknown target f . A small sample residual and controlled readout certify the function without requiring accurate inversion of every coefficient direction. The constants below depend on a specified arithmetic contract, not on an unspecified software library.

E.1 Arithmetic model and the evaluator

Use a binary format with unit roundoff $u \leq 1/4$ and half-minimum-positive-subnormal ν . Round-to-nearest and rounding toward zero satisfy, in the no-overflow range,

$$|\text{RN}(z) - z| \leq u|z| + \nu, \quad |\text{RZ}(z) - z| \leq 2u|z| + 2\nu, \quad |\text{RZ}(z)| \leq |z|. \quad (\text{E.1})$$

Gradual underflow is part of this model; flush-to-zero arithmetic needs a different bound. The stored parameters are finite real floating-point values, and $T_{\text{st}} \geq |\bar{b}| + \sum_j |\bar{w}_j|$ is an upper bound computed with valid rounding.

At a representable input x , the evaluator performs

$$\begin{aligned} \widehat{z}_j &= \text{RN}(\bar{\gamma}_j x + \bar{\beta}_j) \quad \text{in one fused operation,} \\ \widehat{\phi}_j &= \text{RN}(\tanh \widehat{z}_j), \quad \widehat{p}_j = \text{RN}(\bar{w}_j \widehat{\phi}_j), \\ \widehat{q}(x) &= \text{RN}\left(\bar{b} + \sum_j \widehat{p}_j\right), \quad \text{with the sum formed exactly before its final rounding.} \end{aligned} \quad (\text{E.2})$$

Every tanh in this definition is correctly rounded; this is stronger than an unspecified libm accuracy promise. Such semantics are documented for MPFR’s transcendental, fused, and summation operations (MPFR Team, 2025). Its default exponent range and lack of native subnormals must be adjusted or emulated to match a fixed-format contract. Exact accumulation can also be implemented by integer superaccumulators (Neal, 2015). The theorem is about the specified semantics, not a runtime claim for any one implementation.

Lemma E.1 (Uniform activation error with fused arguments). *For every real z for which the operations are in range,*

$$|\text{RN}(\tanh(\text{RN } z)) - \tanh z| \leq 2u + 2\nu. \quad (\text{E.3})$$

Proof. Write $\text{RN } z = z(1 + \theta) + e$ with $|\theta| \leq u$, $|e| \leq \nu$; this follows by clipping the total rounding error to its relative part. Since $\tanh$ is 1-Lipschitz on the real axis, the additive perturbation changes it by at most ν . On the multiplicative segment, the mean value theorem gives

$$|\tanh(z(1 + \theta)) - \tanh z| \leq u|z| \operatorname{sech}^2((1 - u)|z|) \leq \frac{u}{2(1 - u)}.$$

The last inequality follows from $\cosh^2 t = 1 + \sinh^2 t \geq 1 + t^2 \geq 2t$, so $t \operatorname{sech}^2 t \leq 1/2$ for $t \geq 0$. Rounding the final $\tanh$ adds at most $u + \nu$ because its modulus is at most one. For $u \leq 1/4$, the sum is no more than $2u + 2\nu$. $\square$

This lemma is uniform in the slope. Separate rounded multiplication and addition can instead create an absolute argument error proportional to two large cancelling terms, and are not covered by (E.3).

Lemma E.2 (Evaluation of the stored network). *For the evaluator (E.2) at representable core inputs,*

$$|\widehat{q}(x) - q_{\text{st}}(x)| \leq 5uT_{\text{st}} + [2T_{\text{st}} + (1 + u)W + 1]\nu. \quad (\text{E.4})$$

Proof. Correct rounding keeps $\widehat{\phi}_j \in [-1, 1]$. Activation errors contribute at most $(2u + 2\nu) \sum_j |\bar{w}_j|$, and product rounding at most $u \sum_j |\bar{w}_j| + W\nu$. Let $S = \bar{b} + \sum_j \widehat{p}_j$ be the exact accumulated sum. Its magnitude is at most $(1 + u)T_{\text{st}} + W\nu$, so its final rounding costs at most $u(1 + u)T_{\text{st}} + uW\nu + \nu$. Together these are bounded by $(4u + u^2)T_{\text{st}} + [2T_{\text{st}} + (1 + u)W + 1]\nu$, which implies the claim. $\square$

For a different summation scheme replace the final-rounding contribution in this proof by its validated summation bound. Ordinary left-to-right accumulation does not automatically have a width-independent coefficient multiplying u . Exact accumulation removes that source of growth but uses an accumulator wider than a single output-format word.

E.2 Proof of the numerical certificate

Proof of Theorem 3. Apply Lemma D.3 to the exact stored function, using $\Gamma = \max_j |\bar{y}_j|$, the stated $\Gamma h \leq 2\pi/3$, and T_{st} . The target condition $5h/8 < \delta$ follows from shared admissibility as in Appendix D. Since $|f(z_i) - y_i| \leq \eta$ and $\max_i |q_{\text{st}}(z_i) - y_i| \leq r_{\text{cert}}$,

$$\|f - q_{\text{st}}\|_{L^\infty(\Omega)} \leq 2(r_{\text{cert}} + \eta) + 12(B + T_{\text{st}})2^{-\sqrt{s/2}}. \quad (\text{E.5})$$

At representable inputs add (E.4). For a real $x \in [-1, 1]$, let $\bar{x} = \text{RN}(x)$ (clamped to $[-1, 1]$ if necessary). Then $|x - \bar{x}| \leq u + \nu$ and Cauchy's estimate on the target gives $\|f'\|_{L^\infty(\Omega)} \leq B/\delta$. Thus $|f(x) - f(\bar{x})| \leq B(u + \nu)/\delta$. Apply the representable-input certificate at $\bar{x}$ and combine these bounds. This is exactly (9)–(10). $\square$

The input-rounding step compares through f , not through a potentially large derivative bound for the computed candidate. The theorem certifies the actual stored network; agreement with an ideal coefficient vector is not required.

E.3 Enclosing the sample residual

At representable sample nodes, suppose e_i^{eval} bounds $|\widehat{q}(z_i) - q_{\text{st}}(z_i)|$. For stored data y_i , compute $\widehat{r}_i = \text{RN}(\widehat{q}(z_i) - y_i)$. The rounding model gives $|\widehat{q}(z_i) - y_i| \leq (|\widehat{r}_i| + \nu)/(1 - u)$, hence a valid certificate is

$$r_{\text{cert}} = \max_i \left\{ \frac{|\widehat{r}_i| + \nu}{1 - u} + e_i^{\text{eval}} \right\}, \quad (\text{E.6})$$

evaluated with outward rounding. Data evaluation and measurement uncertainty are included in η . For nonrepresentable prescribed nodes, evaluate q_{st} there with interval arithmetic, or account for location rounding separately; one must not silently change the sampling grid used in the theorem.

The coefficient norm T_{st} can be enclosed by exact or outward-rounded summation of the nonnegative magnitudes. The residual and norm checks are valid for an LS output even if the LS solve itself did not impose a coefficient constraint. They are a posteriori statements, not a promise that the checks will pass.

E.4 Assembly, optimization, and storage: a priori output bounds

The posterior certificate is complemented by a perturbation result for the constrained recovery procedure. Let A denote the exact matrix of the prescribed hidden geometry and suppose

$$\max_{i,j} |A_{ij} - \tilde{A}_{ij}| \leq \varepsilon_A. \quad (\text{E.7})$$

Fix $T \geq KB$. Assume the returned vector c is feasible and satisfies

$$\left\| \tilde{A}c - y \right\|_{\infty} \leq \min_{\|v\|_1 \leq T} \left\| \tilde{A}v - y \right\|_{\infty} + \eta_{\text{opt}}. \quad (\text{E.8})$$

Let the stored vector obey $\|\tilde{c}\|_1 \leq T$ and $\|\tilde{c} - c\|_1 \leq \varepsilon_{\text{store}}$.

Proposition E.3 (Perturbed constrained reconstruction). *Under the stated conditions,*

$$\begin{aligned} \|f - q_{\tilde{c}}\|_{\infty} &\leq 2E_{\text{rep}}(W, \lambda) + 4\eta + 4T\varepsilon_A + 2\eta_{\text{opt}} + 2\varepsilon_{\text{store}} \\ &\quad + 12(B + T)2^{-\sqrt{s}/2}. \end{aligned} \quad (\text{E.9})$$

Proof. For every feasible coefficient vector, $\|(A - \tilde{A})c\|_{\infty} \leq \varepsilon_A \|c\|_1 \leq T\varepsilon_A$. Comparison with the explicit feasible witness gives

$$\begin{aligned} \|Ac - y\|_{\infty} &\leq T\varepsilon_A + \left\| \tilde{A}c - y \right\|_{\infty} \\ &\leq T\varepsilon_A + \left\| \tilde{A}c - y \right\|_{\infty} + \eta_{\text{opt}} \\ &\leq E_{\text{rep}}(W, \lambda) + \eta + 2T\varepsilon_A + \eta_{\text{opt}}. \end{aligned}$$

Since the entries of A are bounded by one, $\|A\tilde{c} - Ac\|_{\infty} \leq \varepsilon_{\text{store}}$. Add the data error and apply Lemma D.3 to $q_{\tilde{c}}$. $\square$

No factor $\|A^\dagger\|$ or $\|(A^\top A)^{-1}\|$ occurs in this argument. This is stability of the constrained residual objective and represented function, not stability of every coefficient direction in the raw feature matrix.

Lemma E.4 (A checkable objective gap). *For every v with $\|v\|_1 \leq 1$,*

$$L(v) := y^\top v - T \left\| \tilde{A}^\top v \right\|_{\infty} \leq \min_{\|c\|_1 \leq T} \left\| \tilde{A}c - y \right\|_{\infty}. \quad (\text{E.10})$$

Thus a feasible candidate with $\left\| \tilde{A}c - y \right\|_{\infty} - L(v) \leq \eta_{\text{opt}}$ satisfies (E.8).

Proof. For every feasible c , $y^\top v = v^\top (y - \tilde{A}c) + c^\top \tilde{A}^\top v \leq \|y - \tilde{A}c\|_{\infty} + T \left\| \tilde{A}^\top v \right\|_{\infty}$. Take the infimum. Only weak duality is needed for the certificate. $\square$

E.5 An exact-geometry fixed-format specialization

A convenient sufficient storage design is

$$N = 2^k, \quad \lambda = 1/4, \quad s = 2^\ell, \quad \gamma = N/8, \quad \beta_j = N/8 - j/4, \quad z_i = -1 + 2i/(Ns). \quad (\text{E.11})$$

Require the numerators and exponents of these dyadic values to fit the format, and no overflow in any evaluated operation. For instance Ns and the relevant integer bias numerators may be required not to exceed the exact-integer range of the significand. The restriction to powers of two changes resolution and sample-count choices by at most fixed factors.

Assembling features with (E.2) gives $\varepsilon_A \leq 2u + 2v$ by Lemma E.1; the bias column is exact. Rounding every recovered coefficient toward zero preserves $\|\tilde{c}\|_1 \leq T$ and gives

$$\varepsilon_{\text{store}} \leq 2uT + 2(W + 1)v. \quad (\text{E.12})$$

Combining Proposition E.3, Lemma E.2, and input rounding yields

$$\begin{aligned} \|f - \tilde{q}\|_\infty &\leq 2E_{\text{rep}}(W, \lambda) + 12(B + T)2^{-\sqrt{s/2}} + 4\eta + 2\eta_{\text{opt}} + 17uT \\ &\quad + [10T + (5 + u)W + 5]v + \frac{B}{\delta}(u + v). \end{aligned} \quad (\text{E.13})$$

The coefficient 17 is the sum of the bounds for two uses of feature assembly, storage, and final evaluation. It is not claimed sharp. Arbitrary hidden geometry may instead be rounded and checked directly with Theorem 3; an a priori comparison to the exact witness would then need an additional geometry perturbation bound.

E.6 Finite recovery without an exact-real optimization oracle

The hypothesis (E.8) is verifiable by Lemma E.4 or may be bypassed by the direct residual certificate. There is also a finite, although inefficient, fallback proving computational existence. Choose a rational certified cap $T \geq KB$ and let the assembled entries and labels be rational floating-point numbers. Split $c = c^+ - c^-$ and solve the rational linear program

$$\begin{aligned} \min t, \quad & -t\mathbf{1} \leq \tilde{A}(c^+ - c^-) - y \leq t\mathbf{1}, \\ c^+, c^- \geq 0, \quad & \mathbf{1}^\top(c^+ + c^-) \leq T, \quad 0 \leq t \leq \|y\|_\infty. \end{aligned} \quad (\text{E.14})$$

Its feasible set is nonempty and compact: the zero coefficient vector and $t = \|y\|_\infty$ are feasible. A linear objective attains its optimum at a vertex of this rational polytope. Enumerate the finite collections of independent active constraints, solve each candidate in rational arithmetic, and retain the best feasible vertex. This terminates and gives (E.8) with $\eta_{\text{opt}} = 0$. Rounding the readout toward zero and evaluating as above then gives (E.13). A certified rational bound for K can be formed from (B.23), without truncating an unknown target series.

This fallback uses exact rational working arithmetic; it is not a claim of efficient bit complexity or an all-binary64 optimizer. An efficient numerical LP solver with a checked gap, or an LS candidate passing Theorem 3, is the practical alternative. Working precision can exceed the storage and inference precision. The theorem does not require numerical recovery of the complex continuation or of the original contour coefficients.

E.7 The fixed-floor budget and logarithmic sufficient width

Let $C_{\text{rate}} = 256\pi/(\lambda\delta)$ and fix an admissible W_0 . Set $\tau_a = 128\pi e^{-\pi^2/\lambda}/(\lambda\delta W_0)$. For the constrained candidate, $T = K_{\text{run}}B$ with a certified cap $K_{\text{run}} \geq K$, and (E.9) gives the combined budget

$$\begin{aligned} \|f - \widehat{q}_W\|_\infty &\leq C_{\text{rate}} B e^{-c_{\text{app}} W} + \tau_{\text{tot}}(W), \\ \tau_{\text{tot}}(W) &:= \underbrace{2B\tau_a}_{\text{replica budget}} + \underbrace{12(B+T)2^{-\sqrt{s}/2}}_{\text{sampling}} + \underbrace{4\eta}_{\text{data}} \\ &\quad + \underbrace{4T\varepsilon_A + 2\eta_{\text{opt}} + 2\varepsilon_{\text{store}} + E_{\text{eval}}(W) + B(u+\nu)/\delta}_{\text{certified computation}}. \end{aligned} \tag{E.15}$$

Here E_{eval} is the representable-input bound (E.4). All terms have a defined source. If these quantities are bounded by a common τ throughout an admissible format/range of widths, then (11) follows by solving the first line for W , in addition to $W \geq W_0$. In particular, setting normalized approximation and sampling budgets to $O(u)$, data and optimization error to $O(Bu)$, and keeping the explicit underflow budget at $O(Bu)$ gives

$$\|f - \widehat{q}_W\|_\infty \leq C_{\text{rate}} B e^{-c_{\text{app}} W} + C_{\text{fp}} Bu. \tag{E.16}$$

The constants depend on the selected fixed design and arithmetic contract. Terms proportional to $W\nu$ have not disappeared; they must be bounded over the format's representability and no-overflow range. Refining the requested floor changes λ , the sample multiplier, the coefficient bound, and possibly working precision, so (E.16) is not a joint arbitrary-accuracy complexity theorem.

For an unrestricted LS candidate, use the posterior budget in Theorem 3 instead. Geometric-to-floor behavior is certified when its validated residual follows the approximation envelope and its measured readout norm stays controlled. Neither property is assumed merely because an LS solver was called.

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